

Integrative analysis for sc-/sn-RNA-Seq Analysis in 10X Genomics data

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```
## The following objects are masked _by_ .GlobalEnv:
##
##   PCA_Feature_List_value, Regress_Cell_Cycle_value, assay,
##   genome_name, integration_method, nfeatures_value, root_dir
```

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DNB Bioinformatics Core Pipeline: Standard sc-/sn-RNA-Seq Analysis in 10X Genomics data

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1 Information about this notebook

Biological heterogeneity in single-cell RNA-seq data is often confounded by technical factors including sequencing depth. The number of molecules detected in each cell can vary significantly between cells, even within the same celltype. Interpretation of scRNA-seq data requires effective pre-processing and normalization to remove this technical variability (for more information, see also scRNA-seq Dataset Integration²).

Hence, integration of single-cell sequencing datasets, for example across experimental batches, donors, or conditions, is often an important step in scRNA-seq workflows. Integrative analysis can help to match shared cell types and states across datasets, which can boost statistical power, and most importantly, facilitate accurate comparative analysis across datasets. Here, we include the following integration methods for integrative analysis:

- (1) Seurat ‘anchor-based’³ integration workflow by Choudhary & Satija, 2022⁴,
- (2) Harmony⁵ integration workflow by Korsunsky et al., 2019⁶, and
- (3) Liger⁷ integration workflow by Welch et al., 2019⁸.

Different integration algorithms, such as Seurat, Harmony, and Liger, often use different numbers of dimensions. The optimal number of dimensions depends on the specific algorithm and its associated parameters. While these algorithms generally consider similar variables, the mathematical approaches they use to process and transform those variables differ significantly, which means the same number of dimensions (or “resolutions”) doesn’t work equally well across all algorithms.

Even though they all refer to “resolution,” the effect of resolution can vary from one algorithm to another, and the maximum number of dimensions they can effectively use may differ. Based on our experience, we recommend using 30 dimensions for Seurat, 30 for Harmony, and 20 for Liger. These settings tend to produce the most consistent integration results without unnecessarily increasing computational load.

2 Set up

```
attach(params)
```

```
## The following objects are masked _by_ .GlobalEnv:  
##  
##   PCA_Feature_List_value, Regress_Cell_Cycle_value, assay,  
##   genome_name, integration_method, nfeatures_value, root_dir
```

```
suppressPackageStartupMessages({  
  library(future)  
  library(tidyverse)  
  library(patchwork)  
  library(Seurat)  
  library(SeuratObject)  
  library(harmony)
```

²<https://www.singlecellcourse.org/scrna-seq-dataset-integration.html#seurat-v3-3-vs-5-10k-pbmc>

³https://satijalab.org/seurat/articles/seurat5_integration

⁴<https://doi.org/10.1186/s13059-021-02584-9>

⁵<https://portals.broadinstitute.org/harmony/>

⁶<https://www.nature.com/articles/s41592-019-0619-0>

⁷https://welch-lab.github.io/liger/articles/liger_with_seurat.html

⁸<https://doi.org/10.1016/j.cell.2019.05.006>

```

library(rliger)
library(RcppPlanc)
library(SeuratWrappers)
library(scooter)
library(reshape2)
library(RColorBrewer)
library(knitr)

# Evaluate Seurat R expressions asynchronously when possible using future package
options(future.globals.maxSize = future_globals_value)
plan(multisession, workers = parallelly::availableCores())}

```

3 Directories and paths to file Inputs/Outputs

```

#analysis_dir <- file.path(root_dir, "analyses", "integrative-analysis")
#data_dir <- file.path(root_dir, "analyses", "upstream-analysis", "results", "04_Filter_obj")
#upstream_analysis_dir <- file.path(root_dir, "analyses", "upstream-analysis")

# Input files
#data_file <- file.path(data_dir, "seurat_obj_merged_filtered.rds")

# Create results_dir
#results_dir <- file.path(analysis_dir, "results")
#if (!dir.exists(results_dir)) {
#  dir.create(results_dir)}

# Create plots directory
#report_dir <- file.path(analysis_dir, "plots")
#if (!dir.exists(report_dir)) {
#  dir.create(report_dir)}

#plots_dir <- file.path(report_dir, glue::glue("plots_{integration_method}"))
#if (!dir.exists(plots_dir)) {
#  dir.create(plots_dir)}

source(paste0(root_dir, "/figures/scripts/theme_plot.R"))
#source(paste0(analysis_dir, "/util/function-samples-integrate.R"))
#source(paste0(analysis_dir, "/util/custom-seurat-functions.R"))
#source(paste0(upstream_analysis_dir, "/util/function-process-Seurat.R"))

source(paste0(module_dir, "/util/function-samples-integrate.R"))
source(paste0(module_dir, "/util/custom-seurat-functions.R"))
source(paste0(module_dir, "/util/function-process-Seurat.R"))

```

4 Read seurat object

First, we will use the `seurat_obj_merged_filtered.rds` object as generated from the pipeline in the `upstream-analysis` module. This object contains all samples of the project merged.

```
merged_obj <- readRDS(data_file)
```

5 Seurat ‘anchor-based’ integration workflow

Seurat integration creates a unified object that contains both original data (‘RNA’ assay) as well as integrated data (‘integrated’ assay).

5.1 Normal-sized datasets

SCT method (or `sctransform`) is a modeling framework for the normalization and variance stabilization of molecular count data from scRNA-seq experiments. This procedure omits the need for heuristic steps including pseudocount addition or log-transformation and improves common downstream analytical tasks such as variable gene selection, dimensional reduction, and differential expression. More specifically, the `sctransform` approach utilizes the Pearson residuals from negative binomial regression as input to standard dimensional reduction techniques.

5.2 Large-sized datasets

For very large datasets, the standard integration workflow can sometimes be prohibitively computationally expensive. In addition, Seurat and a lot of integration algorithms are designed to integrate data up to a maximum amount. Here, we use the `big_data` parameter as an option for integration of extremely large datasets. For example, 1.3 millions cells are too much and so this option exist for those datasets. The problem comes down to infinite matrix sizes and memory issues, no computer to data can compute this many variables so new algorithms exist to get around that through various methods. For more information, see Seurat v4 large data integration⁹ and Seurat v5 large data integration¹⁰.

A `reference_list`¹¹ can be used when conducting Seurat integration which help with speeding up integration.

Moreover, we improve efficiency and runtimes by employing (1) Reciprocal PCA (RPCA) and (2) Reference-based integration. The main efficiency improvements are gained in `FindIntegrationAnchors()`. First, we use reciprocal PCA (RPCA), to identify an effective space in which to find anchors. When determining anchors between any two datasets using reciprocal PCA, we project each dataset into the others PCA space and constrain the anchors by the same mutual neighborhood requirement. All downstream integration steps remain the same and we are able to ‘correct’ (or harmonize) the datasets.

Additionally, we use reference-based integration. In the standard workflow, we identify anchors between all pairs of datasets. While this gives datasets equal weight in downstream integration, it can also become computationally intensive. For example when integrating 10 different datasets, we perform 45 different pairwise comparisons. As an alternative, we introduce here the possibility of specifying one or more of the datasets as the ‘reference’ for integrated analysis, with the remainder designated as ‘query’ datasets. In this workflow, we do not identify anchors between pairs of query datasets, reducing the number of comparisons. For example, when integrating 10 datasets with one specified as a reference, we perform only 9 comparisons. Reference-based integration can be applied to either log-normalized or SCTransform-normalized datasets.

Note that the anchor-based RPCA integration represents a faster and more conservative (less correction) method for integration.

⁹https://satijalab.org/seurat/archive/v4.3/integration_large_datasets

¹⁰https://satijalab.org/seurat/articles/seurat5_sketch_analysis

¹¹<https://satijalab.org/seurat/reference/findintegrationanchors#examples>

```

# We will use the Seurat 'anchor-based' workflow, if defined in `params`.
if (use_seurat_integration == "YES"){
  print_message <- "we use the Seurat 'anchor-based' workflow"

  # Make seurat_list
  seurat_obj_list <- list()
  seurat_obj_list <- SplitObject(merged_obj, split.by = "orig.ident")

  # Run function
  seurat_obj <- seurat_integration(seurat_obj_list = seurat_obj_list,
                                   nfeatures_value = nfeatures_value,
                                   num_dim_seurat = num_dim_seurat,
                                   num_dim_seurat_integration = num_dim_seurat_integration,
                                   big_data = big_data_value,
                                   Genome = genome_name,
                                   Regress_Cell_Cycle = Regress_Cell_Cycle_value,
                                   reference_list = reference_list_value,
                                   PCA_Feature_List = PCA_Feature_List_value)
} else {
  print_message <- "we will skip usage of the Seurat 'anchor-based' workflow"
}

```

Here, we will skip usage of the Seurat 'anchor-based' workflow.

6 Harmony integration workflow

Harmony is a general-purpose R package with an efficient algorithm for integrating multiple data sets. It is especially useful for large single-cell datasets such as single-cell RNA-seq. Harmony algorithm projects cells into a shared embedding in which cells group by cell type rather than dataset-specific conditions. Harmony simultaneously accounts for multiple experimental and biological factors. We will run Harmony within our Seurat workflow.

```

# We will use the Harmony integration workflow, if defined in `params`.
if (use_harmony_integration == "YES"){
  print_message <- "we use the Harmony integration workflow"
  seurat_obj <- harmony_integration(seurat_obj = merged_obj,
                                    variables_to_integrate = variable_value,
                                    num_dim = num_dim_harmony)

  # Plot
  name <- paste0(plots_dir, "/", glue::glue("plot_{integration_method}.png"))
  p1 <- DimPlot(object = seurat_obj, reduction = glue::glue("{integration_method}"), pt.size = 100)
  p2 <- VlnPlot(object = seurat_obj, features = "harmony_1", group.by = "orig.ident", pt.size = 100)
  print(patchwork::wrap_plots(list(p1, p2), nrow = 1))
  ggsave(file = name, width = 12, height = 5, device = "png")
} else {
  print_message <- "we will skip usage of the Harmony integration workflow"
}

```

```
## Performing Harmony Integration
```

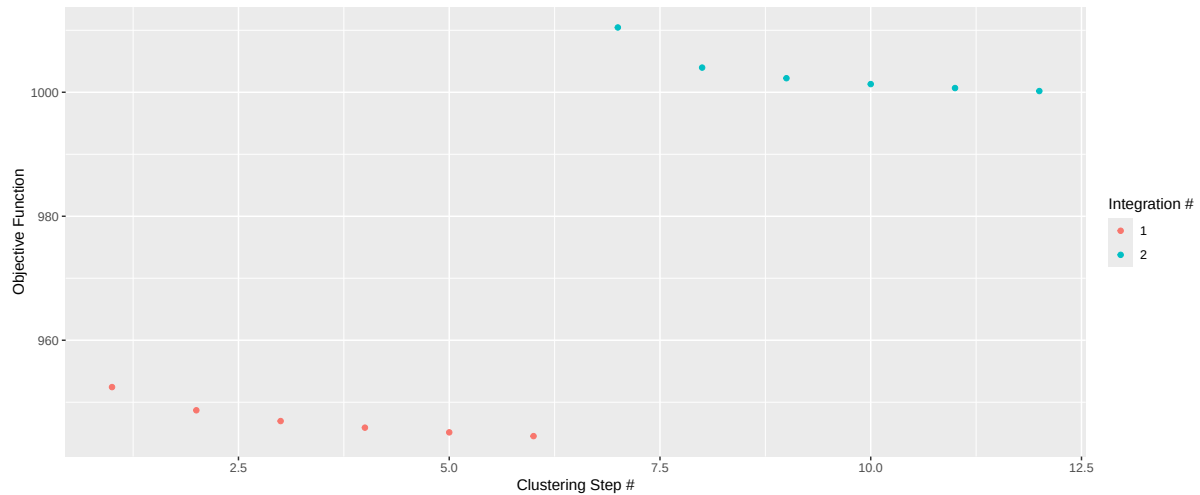
```
## Transposing data matrix
```

```
## Initializing state using k-means centroids initialization
```

```
## Harmony 1/10
```

```
## Harmony 2/10
```

```
## Harmony converged after 2 iterations
```



```
## Performing Uniform Manifold Approximation and Projection (UMAP) dimensional reduction te
```

```
## 19:03:49 UMAP embedding parameters a = 0.9922 b = 1.112
```

```
## 19:03:49 Read 17223 rows and found 30 numeric columns
```

```
## 19:03:49 Using Annoy for neighbor search, n_neighbors = 30
```

```
## 19:03:49 Building Annoy index with metric = cosine, n_trees = 50
```

```
## 0% 10 20 30 40 50 60 70 80 90 100%
```

```
## [----|----|----|----|----|----|----|----|----|----|
```

```
## *****|
```

```
## 19:03:51 Writing NN index file to temp file /lsf_tmp/264625707.tmpdir/RtmpbhtMvE/file2f98
```

```
## 19:03:51 Searching Annoy index using 16 threads, search_k = 3000
```

```
## 19:03:52 Annoy recall = 100%
```

```
## 19:03:53 Commencing smooth kNN distance calibration using 16 threads with target n_neigh
```

```
## 19:03:54 Initializing from normalized Laplacian + noise (using irlba)
```

```
## 19:03:54 Commencing optimization for 200 epochs, with 756636 positive edges
```

```
## 19:04:01 Optimization finished
```

```
## Generate and add metadata
```

```
##
```

```
##
```

```
## ===== saving metadata =====
```

```
##
```

```
##
```

```
##
```

```
##
```

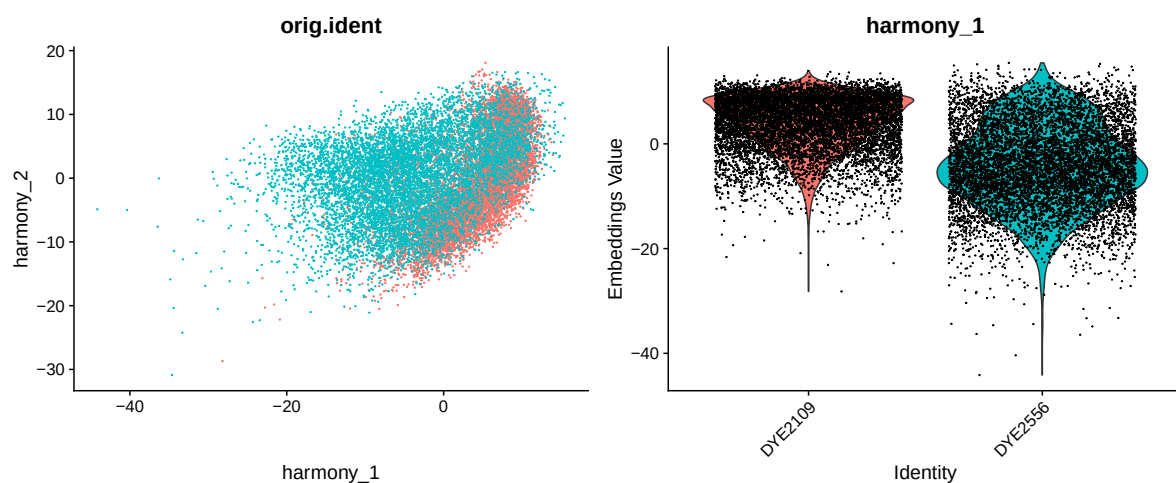
```
## ===== saving metadata =====
```

```
## Computing the k.param nearest neighbors for a given dataset and clusters
```

```
## Computing nearest neighbor graph
```

```
## Computing SNN
```

```
## Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck
##
## Number of nodes: 17223
## Number of edges: 589130
##
## Running Louvain algorithm...
## Maximum modularity in 10 random starts: 0.8112
## Number of communities: 15
## Elapsed time: 2 seconds
## Returned Seurat object
```



Here, we use the Harmony integration workflow.

7 Liger integration workflow

LIGER is an algorithm that enables the identification of shared cell types across individuals, species, and multiple modalities (gene expression, epigenetic or spatial data), as well as dataset-specific features, offering a unified analysis of heterogeneous single-cell datasets.

More specifically, LIGER can be used to integrate datasets: (1) from multiple experiments and biological contexts (differences in technical variation, e.g., number of cells sampled, or in the depth of sequencing allocated to each cell), (2) across space (in situ and single-cell expression data to spatially locate fine subtypes of cells) and (3) various modalities (single-cell RNA-seq and DNA methylation or epigenomes). Across modalities, datasets may vary widely in dynamic range (gene expression versus chromatin accessibility), direction of relationship (RNA-seq versus DNA methylation), or in the number of genes measured (targeted quantification versus unbiased approaches).

```
# We will use the Liger workflow, if defined in `params`.
if (use_liger_integration == "YES"){
  print_message <- "we use the Liger workflow"
  seurat_obj <- liger_integration(seurat_obj = merged_obj,
                                variables_to_integrate = variable_value,
                                n_neighbors = n_neighbors_value)

  # Plot
  name <- paste0(plots_dir, "/", glue::glue("plot_{integration_method}.png"))
  DimPlot(seurat_obj, reduction = glue::glue("{integration_method}"), group.by = c("orig.ident", "DYE2109", "DYE2556"))
}
```



```
ggsave(file = name, width = 6, height = 5, device = "png")

} else {
  print_message <- "we will skip usage of the Liger workflow"}

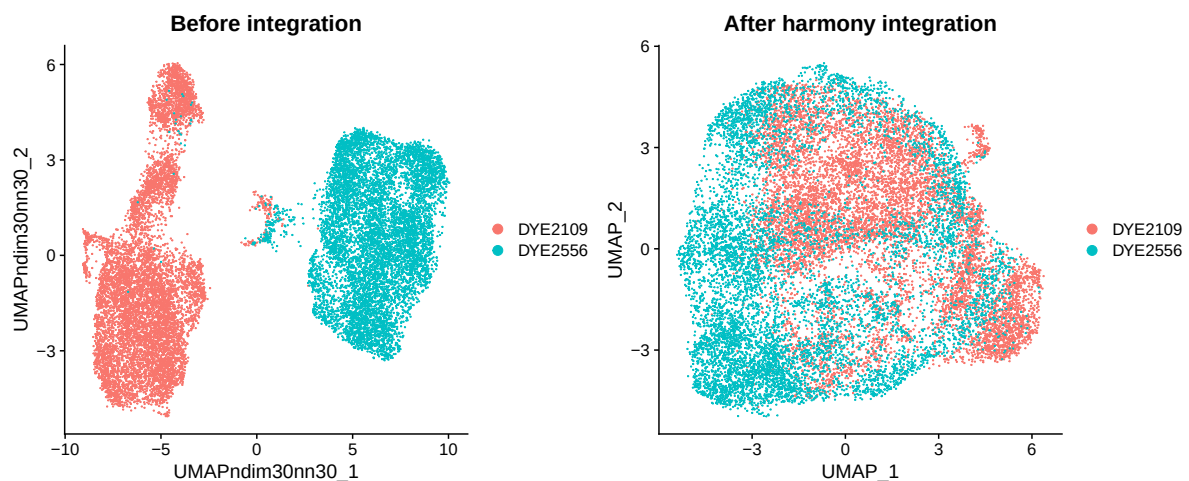
```

Here, we will skip usage of the Liger workflow.

8 Before vs After integration plot

```
# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_unintegrated_vs_{integration_method}.png"))
p1 <- DimPlot(merged_obj, group.by = c("orig.ident")) + ggtitle("Before integration")
p2 <- DimPlot(seurat_obj, reduction = "umap", group.by = c("orig.ident")) + ggtitle(glue::glue("After {integration_method} integration"))
patchwork::wrap_plots(list(p1, p2), nrow = 1)

```



```
ggsave(file = name, width = 12, height = 5, device = "png")

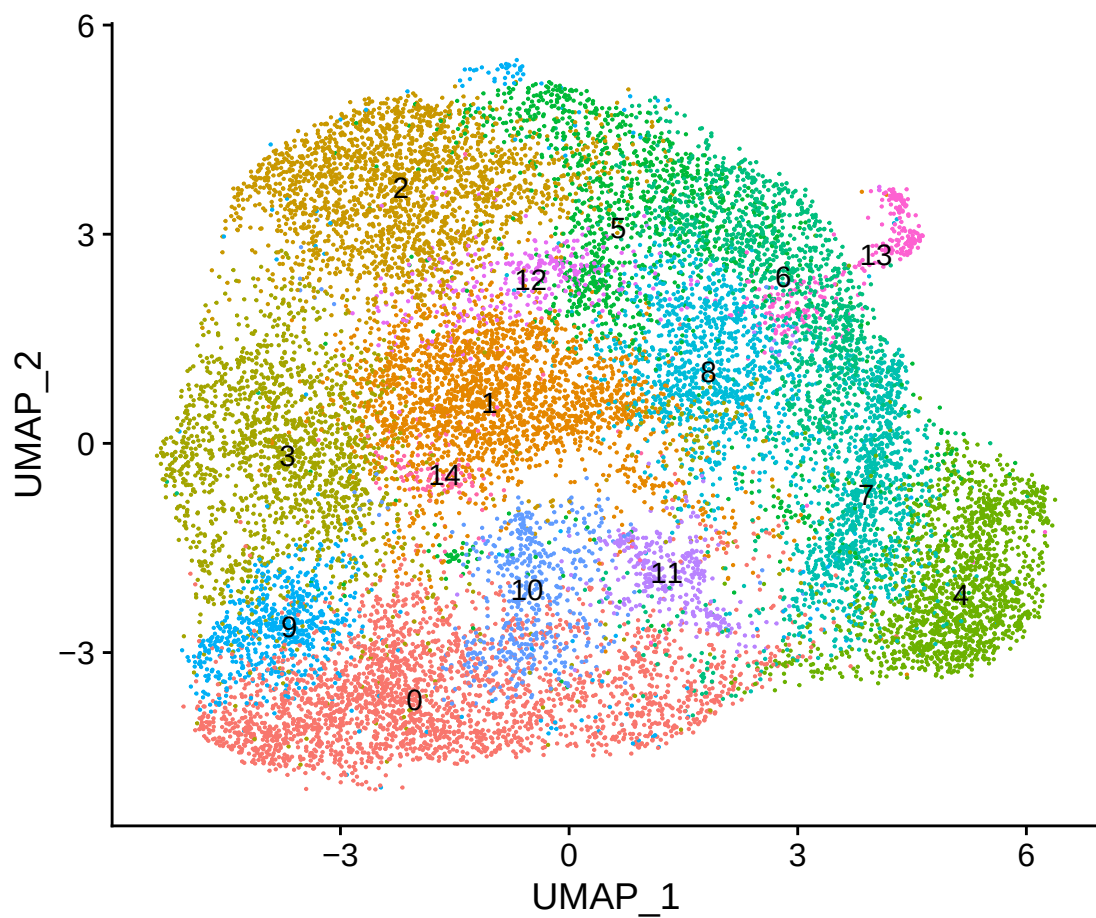
```

```
seurat_obj <- SetIdent(seurat_obj, value = "seurat_clusters")

```

```
# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_clusters_{integration_method}.png"))
DimPlot(seurat_obj, reduction = "umap", label = TRUE) + NoLegend()

```

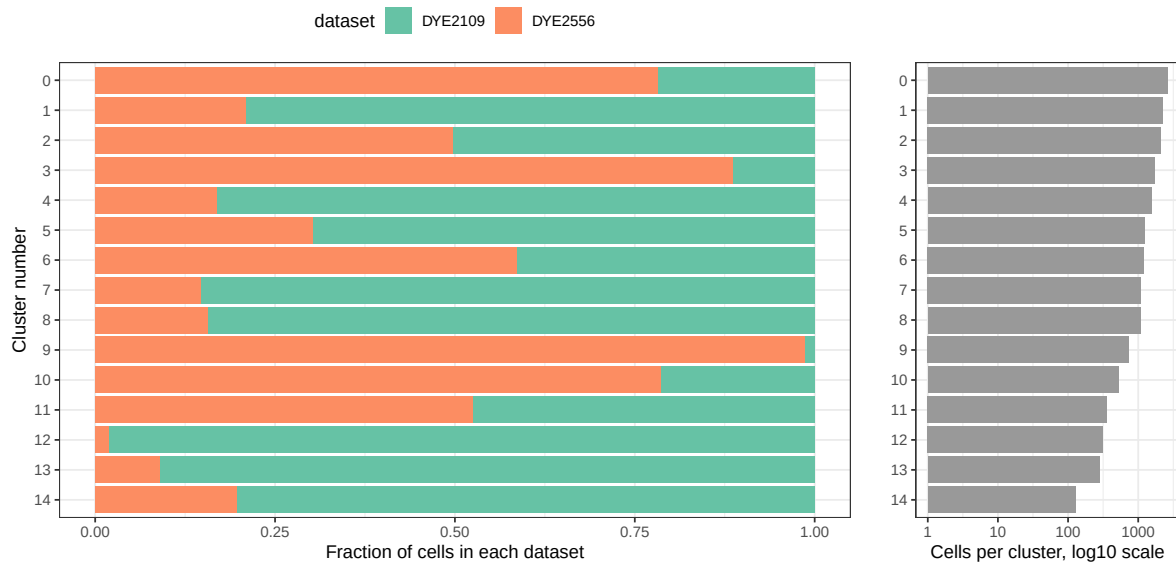


```
ggsave(file = name, width = 6, height = 5, device = "png")
```

```
# Plot
```

```
name <- paste0(plots_dir, "/", glue::glue("plot_fraction_of_cells_{integration_method}.png"),  
plot_integrated_clusters(seurat_obj)
```

```
## Using cluster as id variables
```



```
ggsave(file = name, width = 10, height = 6, device = "png")
```

9 Save output files

```
# Identify columns with a '.1' suffix
cols_to_remove <- grep("\\.1$", colnames(seurat_obj@meta.data), value = TRUE)

# Exclude columns that match the specific patterns (e.g., {assay}_snn_res.0.1, {assay}_snn_res.1.0)
cols_to_remove <- cols_to_remove[!grepl(glue::glue("^.{assay}_snn_res\\.0\\.1$"), cols_to_remove) &&
!grepl(glue::glue("^.{assay}_snn_res\\.1\\.0$"), cols_to_remove) &&
!grepl(glue::glue("^.{assay}_snn_res\\.1\\.10$"), cols_to_remove)]

# Remove the columns
seurat_obj@meta.data <- seurat_obj@meta.data[, !colnames(seurat_obj@meta.data) %in% cols_to_remove]
head(seurat_obj@meta.data)
```

```
##               orig.ident nCount_RNA nFeature_RNA percent.mito
## DYE2109:AAACCCACAACGTGTGT DYE2109         2074         1103         2.555
## DYE2109:AAACCCACAGGACATG DYE2109        18023         4828         3.079
## DYE2109:AAACCCACAGGCGAAT DYE2109        29877        6757         3.330
## DYE2109:AAACCCACATCGCTAA DYE2109         4436         1846         3.291
## DYE2109:AAACCCAGTATCGGTT DYE2109         3300         1473         1.697
## DYE2109:AAACCCAGTCTGTGGC DYE2109        44920        7038         1.823
##               log10GenesPerUMI      ID      SAMPLE
## DYE2109:AAACCCACAACGTGTGT      0.9173202 DYE2109 3149499_DYE2109
## DYE2109:AAACCCACAGGACATG      0.8655820 DYE2109 3149499_DYE2109
## DYE2109:AAACCCACAGGCGAAT      0.8557465 DYE2109 3149499_DYE2109
## DYE2109:AAACCCACATCGCTAA      0.8955962 DYE2109 3149499_DYE2109
## DYE2109:AAACCCAGTATCGGTT      0.9004377 DYE2109 3149499_DYE2109
## DYE2109:AAACCCAGTCTGTGGC      0.8269745 DYE2109 3149499_DYE2109
##
## DYE2109:AAACCCACAACGTGTGT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/
## DYE2109:AAACCCACAGGACATG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/
## DYE2109:AAACCCACAGGCGAAT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/
## DYE2109:AAACCCACATCGCTAA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/
## DYE2109:AAACCCAGTATCGGTT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/
## DYE2109:AAACCCAGTCTGTGGC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/
##               SJUID      SJID      cancer_type
## DYE2109:AAACCCACAACGTGTGT SJHBBBMYJSX SJOS030605 Osteosarcoma
## DYE2109:AAACCCACAGGACATG SJHBBBMYJSX SJOS030605 Osteosarcoma
## DYE2109:AAACCCACAGGCGAAT SJHBBBMYJSX SJOS030605 Osteosarcoma
## DYE2109:AAACCCACATCGCTAA SJHBBBMYJSX SJOS030605 Osteosarcoma
## DYE2109:AAACCCAGTATCGGTT SJHBBBMYJSX SJOS030605 Osteosarcoma
## DYE2109:AAACCCAGTCTGTGGC SJHBBBMYJSX SJOS030605 Osteosarcoma
```

##	Subject_Disease_Code	Subject_Disease_Subtype_Code	
## DYE2109:AAACCCACAACACTGTGT	OST	OST	
## DYE2109:AAACCCACAGGACATG	OST	OST	
## DYE2109:AAACCCACAGGCGAAT	OST	OST	
## DYE2109:AAACCCACATCGCTAA	OST	OST	
## DYE2109:AAACCCAGTATCGGTT	OST	OST	
## DYE2109:AAACCCAGTCTGTGGC	OST	OST	
##	tissue_type	assay_technology_updated	assay_technology
## DYE2109:AAACCCACAACACTGTGT	CCLF_cells	10x_scrNaseq	10x_scrNaseq
## DYE2109:AAACCCACAGGACATG	CCLF_cells	10x_scrNaseq	10x_scrNaseq
## DYE2109:AAACCCACAGGCGAAT	CCLF_cells	10x_scrNaseq	10x_scrNaseq
## DYE2109:AAACCCACATCGCTAA	CCLF_cells	10x_scrNaseq	10x_scrNaseq
## DYE2109:AAACCCAGTATCGGTT	CCLF_cells	10x_scrNaseq	10x_scrNaseq
## DYE2109:AAACCCAGTCTGTGGC	CCLF_cells	10x_scrNaseq	10x_scrNaseq
##	historic_data	RPMI	genome_reference_updated_code
## DYE2109:AAACCCACAACACTGTGT	no	no	human-genome
## DYE2109:AAACCCACAGGACATG	no	no	human-genome
## DYE2109:AAACCCACAGGCGAAT	no	no	human-genome
## DYE2109:AAACCCACATCGCTAA	no	no	human-genome
## DYE2109:AAACCCAGTATCGGTT	no	no	human-genome
## DYE2109:AAACCCAGTCTGTGGC	no	no	human-genome
##	genome_reference_updated		
## DYE2109:AAACCCACAACACTGTGT	human-genome (tested for mouse cells: None)		
## DYE2109:AAACCCACAGGACATG	human-genome (tested for mouse cells: None)		
## DYE2109:AAACCCACAGGCGAAT	human-genome (tested for mouse cells: None)		
## DYE2109:AAACCCACATCGCTAA	human-genome (tested for mouse cells: None)		
## DYE2109:AAACCCAGTATCGGTT	human-genome (tested for mouse cells: None)		
## DYE2109:AAACCCAGTCTGTGGC	human-genome (tested for mouse cells: None)		
##	genome_reference		
## DYE2109:AAACCCACAACACTGTGT	dual-genome		
## DYE2109:AAACCCACAGGACATG	dual-genome		
## DYE2109:AAACCCACAGGCGAAT	dual-genome		
## DYE2109:AAACCCACATCGCTAA	dual-genome		
## DYE2109:AAACCCAGTATCGGTT	dual-genome		
## DYE2109:AAACCCAGTCTGTGGC	dual-genome		
##			
## DYE2109:AAACCCACAACACTGTGT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/		
## DYE2109:AAACCCACAGGACATG	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/		
## DYE2109:AAACCCACAGGCGAAT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/		
## DYE2109:AAACCCACATCGCTAA	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/		
## DYE2109:AAACCCAGTATCGGTT	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/		
## DYE2109:AAACCCAGTCTGTGGC	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/		
##	DYExxxx	DYE_Melody_unsure	DYExxxx_corrected
## DYE2109:AAACCCACAACACTGTGT	DYE2109	<NA>	DYE2109
## DYE2109:AAACCCACAGGACATG	DYE2109	<NA>	DYE2109

## DYE2109:AAACCCACAGGCGAAT	DYE2109	<NA>	DYE2109
## DYE2109:AAACCCACATCGCTAA	DYE2109	<NA>	DYE2109
## DYE2109:AAACCCAGTATCGGTT	DYE2109	<NA>	DYE2109
## DYE2109:AAACCCAGTCTGTGGC	DYE2109	<NA>	DYE2109
##	SRM_updated	SRM	SRM_Sample_ID_updated SRM_Sample_ID
## DYE2109:AAACCCACAACGTGTGT	829202	829202	3149499 3149499
## DYE2109:AAACCCACAGGACATG	829202	829202	3149499 3149499
## DYE2109:AAACCCACAGGCGAAT	829202	829202	3149499 3149499
## DYE2109:AAACCCACATCGCTAA	829202	829202	3149499 3149499
## DYE2109:AAACCCAGTATCGGTT	829202	829202	3149499 3149499
## DYE2109:AAACCCAGTCTGTGGC	829202	829202	3149499 3149499
##	Experimental_Sample	sample_3D	sample_2D Tissue_Bank_ID
## DYE2109:AAACCCACAACGTGTGT	SJ1065	No	No TB-18-
02891			
## DYE2109:AAACCCACAGGACATG	SJ1065	No	No TB-18-
02891			
## DYE2109:AAACCCACAGGCGAAT	SJ1065	No	No TB-18-
02891			
## DYE2109:AAACCCACATCGCTAA	SJ1065	No	No TB-18-
02891			
## DYE2109:AAACCCAGTATCGGTT	SJ1065	No	No TB-18-
02891			
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02891			
##	Xenograft_CompBio_ID	CSTN_MAST_ID	Date
## DYE2109:AAACCCACAACGTGTGT	SJOS030605_X2	MAST413B	2024-06-14T00:00:00Z
## DYE2109:AAACCCACAGGACATG	SJOS030605_X2	MAST413B	2024-06-14T00:00:00Z
## DYE2109:AAACCCACAGGCGAAT	SJOS030605_X2	MAST413B	2024-06-14T00:00:00Z
## DYE2109:AAACCCACATCGCTAA	SJOS030605_X2	MAST413B	2024-06-14T00:00:00Z
## DYE2109:AAACCCAGTATCGGTT	SJOS030605_X2	MAST413B	2024-06-14T00:00:00Z
## DYE2109:AAACCCAGTCTGTGGC	SJOS030605_X2	MAST413B	2024-06-14T00:00:00Z
##	SRM_Submission_Information		
## DYE2109:AAACCCACAACGTGTGT	<NA>		
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## DYE2109:AAACCCAGTCTGTGGC	<NA>		
##	Compbioname_for_publications	Comp_Bio_Sample_Name	
## DYE2109:AAACCCACAACGTGTGT	SJOS030605_C3	SJOS030605_C3-SJ1065T_2	
## DYE2109:AAACCCACAGGACATG	SJOS030605_C3	SJOS030605_C3-SJ1065T_2	
## DYE2109:AAACCCACAGGCGAAT	SJOS030605_C3	SJOS030605_C3-SJ1065T_2	
## DYE2109:AAACCCACATCGCTAA	SJOS030605_C3	SJOS030605_C3-SJ1065T_2	
## DYE2109:AAACCCAGTATCGGTT	SJOS030605_C3	SJOS030605_C3-SJ1065T_2	
## DYE2109:AAACCCAGTCTGTGGC	SJOS030605_C3	SJOS030605_C3-SJ1065T_2	
##	Comp_Bio_Subject_ID	Preferred_Sample_ID	
## DYE2109:AAACCCACAACGTGTGT	SJ030605	SJ1065T_2	
## DYE2109:AAACCCACAGGACATG	SJ030605	SJ1065T_2	
## DYE2109:AAACCCACAGGCGAAT	SJ030605	SJ1065T_2	
## DYE2109:AAACCCACATCGCTAA	SJ030605	SJ1065T_2	
## DYE2109:AAACCCAGTATCGGTT	SJ030605	SJ1065T_2	

##	DYE2109:AAACCCAGTCTGTGGC	SJ030605	SJ1065T_2			
##		Sample_Type_Code	Prior_Comp_Bio_ID	PATH	NOTES	
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##	DYE2109:AAACCCACAGGACATG	C	SJOS030605	<NA>	NA	
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##	DYE2109:AAACCCACATCGCTAA	C	SJOS030605	<NA>	NA	
##	DYE2109:AAACCCAGTATCGGTT	C	SJOS030605	<NA>	NA	
##	DYE2109:AAACCCAGTCTGTGGC	C	SJOS030605	<NA>	NA	
##		match_SJID_RPMI	S.Score	G2M.Score	Phase	
##	DYE2109:AAACCCACAACACTGTGT	SJOS030605_no	0.029309689	-0.143925507	S	
##	DYE2109:AAACCCACAGGACATG	SJOS030605_no	-0.145495825	0.172341128	G2M	
##	DYE2109:AAACCCACAGGCGAAT	SJOS030605_no	0.103113321	0.017577024	S	
##	DYE2109:AAACCCACATCGCTAA	SJOS030605_no	-0.075474864	0.238239019	G2M	
##	DYE2109:AAACCCAGTATCGGTT	SJOS030605_no	-0.011282013	0.006574403	G2M	
##	DYE2109:AAACCCAGTCTGTGGC	SJOS030605_no	0.003546398	0.188333123	G2M	
##		UMAP_1	UMAP_2	PC_1	PC_2	PC_3
##	DYE2109:AAACCCACAACACTGTGT	3.519175	-1.921958	9.663225	7.078574	4.0789877
##	DYE2109:AAACCCACAGGACATG	-1.152715	3.024430	7.195771	-10.952481	-3.0915320
##	DYE2109:AAACCCACAGGCGAAT	-2.008140	2.096559	2.055668	-12.429890	3.2396863
##	DYE2109:AAACCCACATCGCTAA	4.085037	1.523171	12.802924	1.169744	-1.3242469
##	DYE2109:AAACCCAGTATCGGTT	4.003570	-0.404878	13.135035	8.753355	0.2016875
##	DYE2109:AAACCCAGTCTGTGGC	1.378143	3.335749	10.007553	-7.405388	-4.3712622
##		PC_4	PC_5	PC_6	PC_7	
##	DYE2109:AAACCCACAACACTGTGT	-0.8989851	3.4785479	4.6833617	-1.579191056	
##	DYE2109:AAACCCACAGGACATG	2.2747797	1.4447327	2.2701193	-0.277084461	
##	DYE2109:AAACCCACAGGCGAAT	-2.3822854	1.1932545	-2.5911117	1.823911254	
##	DYE2109:AAACCCACATCGCTAA	-2.7640391	1.8122737	-0.9178103	0.009797272	
##	DYE2109:AAACCCAGTATCGGTT	0.5418655	0.2573033	0.4390628	-1.294562778	
##	DYE2109:AAACCCAGTCTGTGGC	-0.8848126	-0.6372359	-3.5383855	1.002155348	
##		PC_8	PC_9	PC_10	PC_11	PC_12
##	DYE2109:AAACCCACAACACTGTGT	-1.0247210	0.81710793	-2.1728923	0.5505248	-
	2.481885					
##	DYE2109:AAACCCACAGGACATG	0.8129334	-6.87471206	-3.9070574	0.4317886	2.132460
##	DYE2109:AAACCCACAGGCGAAT	0.4732263	2.15869300	0.8648879	-2.0360958	3.149628
##	DYE2109:AAACCCACATCGCTAA	-0.9995096	-0.03948597	-2.4479940	0.1144913	1.993451
##	DYE2109:AAACCCAGTATCGGTT	-1.0672499	-0.18567736	1.4473227	0.7199554	-
	0.365992					
##	DYE2109:AAACCCAGTCTGTGGC	-0.8488276	-0.67607856	0.8396755	1.5194082	1.726608
##		PC_13	PC_14	PC_15	PC_16	PC_17
##	DYE2109:AAACCCACAACACTGTGT	-1.9946774	-0.6889168	-0.8610748	-3.5271137	2.2253711
##	DYE2109:AAACCCACAGGACATG	-0.1719568	-1.3127177	0.7189636	1.1575003	-
	0.5746315					
##	DYE2109:AAACCCACAGGCGAAT	-0.7021116	2.2996708	0.3662269	-3.1561115	1.6115634
##	DYE2109:AAACCCACATCGCTAA	-2.9871921	0.3712947	-1.3384600	0.1978398	-
	2.4948900					
##	DYE2109:AAACCCAGTATCGGTT	-0.2702163	1.0059367	1.3182174	-1.3353218	-
	0.1862197					
##	DYE2109:AAACCCAGTCTGTGGC	-0.2499420	-0.3055286	-1.4799158	0.0639429	-
	1.4270683					
##		PC_18	PC_19	PC_20	PC_21	
##	DYE2109:AAACCCACAACACTGTGT	-2.6493056	-1.2865859	1.17429298	-0.17077402	

## DYE2109:AAACCCACAGGACATG	0.2223031	1.7587408	0.47740384	1.28842937
## DYE2109:AAACCCACAGGCGAAT	0.5869538	-1.7702607	-0.54852918	-0.35742542
## DYE2109:AAACCCACATCGCTAA	-0.9557972	-0.5418771	0.30826898	0.42824778
## DYE2109:AAACCCAGTATCGGTT	-1.9391193	0.8590362	1.50789450	-0.83889903
## DYE2109:AAACCCAGTCTGTGGC	0.3998942	-0.4364540	0.05976991	0.03943408
##	PC_22	PC_23	PC_24	PC_25
## DYE2109:AAACCCACAACACTGTGT	-0.4801928	0.3626377	-0.1410548	-2.19606867
## DYE2109:AAACCCACAGGACATG	0.2491494	-0.3377164	-2.2398115	0.39198605
## DYE2109:AAACCCACAGGCGAAT	0.8297882	-1.0335503	-0.8881883	1.27154928
## DYE2109:AAACCCACATCGCTAA	0.2638503	0.5693051	1.6517613	-0.09858024
## DYE2109:AAACCCAGTATCGGTT	0.9235641	0.2380381	1.8289060	0.33907460
## DYE2109:AAACCCAGTCTGTGGC	-2.3471296	0.2552280	2.0036406	-0.46186112
##	PC_26	PC_27	PC_28	PC_29
## DYE2109:AAACCCACAACACTGTGT	-0.79650332	-0.9703837	0.77233203	-0.88477123
## DYE2109:AAACCCACAGGACATG	0.09935344	-2.5395412	0.01735975	0.94930022
## DYE2109:AAACCCACAGGCGAAT	-1.51646604	-0.2892143	-0.57249556	-0.75638887
## DYE2109:AAACCCACATCGCTAA	0.30208300	-3.7138746	1.60064143	0.05072245
## DYE2109:AAACCCAGTATCGGTT	2.09423280	-0.3523753	-1.52177646	-0.62834183
## DYE2109:AAACCCAGTCTGTGGC	1.18240245	-2.3649857	0.83644876	0.29585985
##	PC_30	RNA_snn_res.0	seurat_clusters	
## DYE2109:AAACCCACAACACTGTGT	0.52189970		1	7
## DYE2109:AAACCCACAGGACATG	-0.04705888		1	2
## DYE2109:AAACCCACAGGCGAAT	0.24990605		1	12
## DYE2109:AAACCCACATCGCTAA	1.06225886		1	6
## DYE2109:AAACCCAGTATCGGTT	-0.28853131		1	7
## DYE2109:AAACCCAGTCTGTGGC	2.02818241		1	6
##	RNA_snn_res.0.1	RNA_snn_res.0.2	RNA_snn_res.0.3	
## DYE2109:AAACCCACAACACTGTGT	3	3	4	
## DYE2109:AAACCCACAGGACATG	1	1	3	
## DYE2109:AAACCCACAGGCGAAT	1	1	2	
## DYE2109:AAACCCACATCGCTAA	1	3	4	
## DYE2109:AAACCCAGTATCGGTT	3	3	4	
## DYE2109:AAACCCAGTCTGTGGC	1	3	4	
##	RNA_snn_res.0.4	RNA_snn_res.0.5	RNA_snn_res.0.6	
## DYE2109:AAACCCACAACACTGTGT	7	8	9	
## DYE2109:AAACCCACAGGACATG	3	4	4	
## DYE2109:AAACCCACAGGCGAAT	2	2	2	
## DYE2109:AAACCCACATCGCTAA	7	3	3	
## DYE2109:AAACCCAGTATCGGTT	7	8	9	
## DYE2109:AAACCCAGTCTGTGGC	4	3	3	
##	RNA_snn_res.0.7	RNA_snn_res.0.8	RNA_snn_res.0.9	
## DYE2109:AAACCCACAACACTGTGT	8	7	10	
## DYE2109:AAACCCACAGGACATG	3	2	3	
## DYE2109:AAACCCACAGGCGAAT	5	12	11	
## DYE2109:AAACCCACATCGCTAA	8	6	9	
## DYE2109:AAACCCAGTATCGGTT	8	7	10	
## DYE2109:AAACCCAGTCTGTGGC	5	6	5	
##	RNA_snn_res.1	RNA_snn_res.2	RNA_snn_res.3	
## DYE2109:AAACCCACAACACTGTGT	10	9	5	
## DYE2109:AAACCCACAGGACATG	2	1	4	
## DYE2109:AAACCCACAGGCGAAT	11	17	20	

## DYE2109:AAACCCACATCGCTAA	9	13	28
## DYE2109:AAACCCAGTATCGGTT	10	9	5
## DYE2109:AAACCCAGTCTGTGGC	6	14	19
##	RNA_snn_res.4	RNA_snn_res.5	RNA_snn_res.6
## DYE2109:AAACCCACAACGTGTGT	24	22	18
## DYE2109:AAACCCACAGGACATG	31	26	20
## DYE2109:AAACCCACAGGCGAAT	20	34	37
## DYE2109:AAACCCACATCGCTAA	34	31	22
## DYE2109:AAACCCAGTATCGGTT	23	16	10
## DYE2109:AAACCCAGTCTGTGGC	25	13	19
##	RNA_snn_res.7	RNA_snn_res.8	RNA_snn_res.9
## DYE2109:AAACCCACAACGTGTGT	14	13	12
## DYE2109:AAACCCACAGGACATG	26	24	19
## DYE2109:AAACCCACAGGCGAAT	35	43	34
## DYE2109:AAACCCACATCGCTAA	19	65	65
## DYE2109:AAACCCAGTATCGGTT	9	6	13
## DYE2109:AAACCCAGTCTGTGGC	38	48	47
##	RNA_snn_res.10	orig.ident.1	nCount_RNA.1
## DYE2109:AAACCCACAACGTGTGT	4	DYE2109	2074
## DYE2109:AAACCCACAGGACATG	17	DYE2109	18023
## DYE2109:AAACCCACAGGCGAAT	34	DYE2109	29877
## DYE2109:AAACCCACATCGCTAA	67	DYE2109	4436
## DYE2109:AAACCCAGTATCGGTT	7	DYE2109	3300
## DYE2109:AAACCCAGTCTGTGGC	51	DYE2109	44920
##	nFeature_RNA.1	percent.mito.1	log10GenesPerUMI.1
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## DYE2109:AAACCCACAGGACATG	4828	3.079	0.8655820
## DYE2109:AAACCCACAGGCGAAT	6757	3.330	0.8557465
## DYE2109:AAACCCACATCGCTAA	1846	3.291	0.8955962
## DYE2109:AAACCCAGTATCGGTT	1473	1.697	0.9004377
## DYE2109:AAACCCAGTCTGTGGC	7038	1.823	0.8269745
##	ID.1	SAMPLE.1	
## DYE2109:AAACCCACAACGTGTGT	DYE2109	3149499_DYE2109	
## DYE2109:AAACCCACAGGACATG	DYE2109	3149499_DYE2109	
## DYE2109:AAACCCACAGGCGAAT	DYE2109	3149499_DYE2109	
## DYE2109:AAACCCACATCGCTAA	DYE2109	3149499_DYE2109	
## DYE2109:AAACCCAGTATCGGTT	DYE2109	3149499_DYE2109	
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## DYE2109:AAACCCACAGGACATG	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/		
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## DYE2109:AAACCCAGTCTGTGGC	/research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina/1/3149499/		


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##          SJUID.1      SJID.1 cancer_type.1
## DYE2109:AAACCCACAACGTGTGT SJHBBBMYJSX SJOS030605 Osteosarcoma
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## DYE2109:AAACCCACAGGCGAAT SJHBBBMYJSX SJOS030605 Osteosarcoma
## DYE2109:AAACCCACATCGCTAA SJHBBBMYJSX SJOS030605 Osteosarcoma
## DYE2109:AAACCCAGTATCGGTT SJHBBBMYJSX SJOS030605 Osteosarcoma
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##          Subject_Disease_Code.1 Subject_Disease_Subtype_Code.1
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## DYE2109:AAACCCACAACGTGTGT CCLF_cells 10x_scRNAseq
## DYE2109:AAACCCACAGGACATG CCLF_cells 10x_scRNAseq
## DYE2109:AAACCCACAGGCGAAT CCLF_cells 10x_scRNAseq
## DYE2109:AAACCCACATCGCTAA CCLF_cells 10x_scRNAseq
## DYE2109:AAACCCAGTATCGGTT CCLF_cells 10x_scRNAseq
## DYE2109:AAACCCAGTCTGTGGC CCLF_cells 10x_scRNAseq
##          assay_technology.1 historic_data.1 RPMI.1
## DYE2109:AAACCCACAACGTGTGT 10x_scRNAseq no no
## DYE2109:AAACCCACAGGACATG 10x_scRNAseq no no
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## DYE2109:AAACCCACATCGCTAA 10x_scRNAseq no no
## DYE2109:AAACCCAGTATCGGTT 10x_scRNAseq no no
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##          genome_reference_updated_code.1
## DYE2109:AAACCCACAACGTGTGT human-genome
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##
## DYE2109:AAACCCACAACGTGTGT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumina

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1/3149499/
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## DYE2109:AAACCCACAGGCGAAT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
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## DYE2109:AAACCCAGTCTGTGGC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illum
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##
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## DYE2109:AAACCCACAACGTGTGT 829202 829202 3149499
## DYE2109:AAACCCACAGGACATG 829202 829202 3149499
## DYE2109:AAACCCACAGGCGAAT 829202 829202 3149499
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## DYE2109:AAACCCAGTCTGTGGC 829202 829202 3149499
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SRM_Sample_ID.1 Experimental_Sample.1 sample_3D.1
## DYE2109:AAACCCACAACGTGTGT 3149499 SJ1065 No
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## DYE2109:AAACCCACAGGCGAAT 3149499 SJ1065 No
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sample_2D.1 Tissue_Bank_ID.1 Xenograft_CompBio_ID.1
## DYE2109:AAACCCACAACGTGTGT No TB-18-02891 SJOS030605_X2
## DYE2109:AAACCCACAGGACATG No TB-18-02891 SJOS030605_X2
## DYE2109:AAACCCACAGGCGAAT No TB-18-02891 SJOS030605_X2
## DYE2109:AAACCCACATCGCTAA No TB-18-02891 SJOS030605_X2
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##
CSTN_MAST_ID.1 Date.1
## DYE2109:AAACCCACAACGTGTGT MAST413B 2024-06-14T00:00:00Z
## DYE2109:AAACCCACAGGACATG MAST413B 2024-06-14T00:00:00Z
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SRM_Submission_Information.1
## DYE2109:AAACCCACAACGTGTGT <NA>
## DYE2109:AAACCCACAGGACATG <NA>
## DYE2109:AAACCCACAGGCGAAT <NA>
## DYE2109:AAACCCACATCGCTAA <NA>

```

##	DYE2109:AAACCCAGTATCGGTT	<NA>				
##	DYE2109:AAACCCAGTCTGTGGC	<NA>				
##		Compbioname_for_publications.1	Comp_Bio_Sample_Name.1			
##	DYE2109:AAACCCACAACGTGTGT	SJOS030605_C3	SJOS030605_C3-			
##	DYE2109:AAACCCACAGGACATG	SJOS030605_C3	SJOS030605_C3-			
##	DYE2109:AAACCCACAGGCGAAT	SJOS030605_C3	SJOS030605_C3-			
##	DYE2109:AAACCCACATCGCTAA	SJOS030605_C3	SJOS030605_C3-			
##	DYE2109:AAACCCAGTATCGGTT	SJOS030605_C3	SJOS030605_C3-			
##	DYE2109:AAACCCAGTCTGTGGC	SJOS030605_C3	SJOS030605_C3-			
##		Comp_Bio_Subject_ID.1	Preferred_Sample_ID.1			
##	DYE2109:AAACCCACAACGTGTGT	SJ030605	SJ1065T_2			
##	DYE2109:AAACCCACAGGACATG	SJ030605	SJ1065T_2			
##	DYE2109:AAACCCACAGGCGAAT	SJ030605	SJ1065T_2			
##	DYE2109:AAACCCACATCGCTAA	SJ030605	SJ1065T_2			
##	DYE2109:AAACCCAGTATCGGTT	SJ030605	SJ1065T_2			
##	DYE2109:AAACCCAGTCTGTGGC	SJ030605	SJ1065T_2			
##		Sample_Type_Code.1	Prior_Comp_Bio_ID.1	PATH.1	NOTES.1	
##	DYE2109:AAACCCACAACGTGTGT	C	SJOS030605	<NA>	NA	
##	DYE2109:AAACCCACAGGACATG	C	SJOS030605	<NA>	NA	
##	DYE2109:AAACCCACAGGCGAAT	C	SJOS030605	<NA>	NA	
##	DYE2109:AAACCCACATCGCTAA	C	SJOS030605	<NA>	NA	
##	DYE2109:AAACCCAGTATCGGTT	C	SJOS030605	<NA>	NA	
##	DYE2109:AAACCCAGTCTGTGGC	C	SJOS030605	<NA>	NA	
##		match_SJID_RPMI.1	S.Score.1	G2M.Score.1	Phase.1	
##	DYE2109:AAACCCACAACGTGTGT	SJOS030605_no	0.029309689	-0.143925507	S	
##	DYE2109:AAACCCACAGGACATG	SJOS030605_no	-0.145495825	0.172341128	G2M	
##	DYE2109:AAACCCACAGGCGAAT	SJOS030605_no	0.103113321	0.017577024	S	
##	DYE2109:AAACCCACATCGCTAA	SJOS030605_no	-0.075474864	0.238239019	G2M	
##	DYE2109:AAACCCAGTATCGGTT	SJOS030605_no	-0.011282013	0.006574403	G2M	
##	DYE2109:AAACCCAGTCTGTGGC	SJOS030605_no	0.003546398	0.188333123	G2M	
##		UMAP_1.x	UMAP_2.x	PC_1.x	PC_2.x	PC_3.x
##	DYE2109:AAACCCACAACGTGTGT	3.519175	-1.921958	9.663225	7.078574	4.0789877
##	DYE2109:AAACCCACAGGACATG	-1.152715	3.024430	7.195771	-10.952481	-3.0915320
##	DYE2109:AAACCCACAGGCGAAT	-2.008140	2.096559	2.055668	-12.429890	3.2396863
##	DYE2109:AAACCCACATCGCTAA	4.085037	1.523171	12.802924	1.169744	-1.3242469
##	DYE2109:AAACCCAGTATCGGTT	4.003570	-0.404878	13.135035	8.753355	0.2016875
##	DYE2109:AAACCCAGTCTGTGGC	1.378143	3.335749	10.007553	-7.405388	-4.3712622
##		PC_4.x	PC_5.x	PC_6.x	PC_7.x	
##	DYE2109:AAACCCACAACGTGTGT	-0.8989851	3.4785479	4.6833617	-1.579191056	
##	DYE2109:AAACCCACAGGACATG	2.2747797	1.4447327	2.2701193	-0.277084461	
##	DYE2109:AAACCCACAGGCGAAT	-2.3822854	1.1932545	-2.5911117	1.823911254	
##	DYE2109:AAACCCACATCGCTAA	-2.7640391	1.8122737	-0.9178103	0.009797272	
##	DYE2109:AAACCCAGTATCGGTT	0.5418655	0.2573033	0.4390628	-1.294562778	
##	DYE2109:AAACCCAGTCTGTGGC	-0.8848126	-0.6372359	-3.5383855	1.002155348	
##		PC_8.x	PC_9.x	PC_10.x	PC_11.x	PC_12.x

## DYE2109:AAACCCACA	ACTGTGT	-1.0247210	0.81710793	-2.1728923	0.5505248	-
2.481885						
## DYE2109:AAACCCACAGGACATG		0.8129334	-6.87471206	-3.9070574	0.4317886	2.132460
## DYE2109:AAACCCACAGGCGAAT		0.4732263	2.15869300	0.8648879	-2.0360958	3.149628
## DYE2109:AAACCCACATCGCTAA		-0.9995096	-0.03948597	-2.4479940	0.1144913	1.993451
## DYE2109:AAACCCAGTATCGGTT		-1.0672499	-0.18567736	1.4473227	0.7199554	-
0.365992						
## DYE2109:AAACCCAGTCTGTGGC		-0.8488276	-0.67607856	0.8396755	1.5194082	1.726608
##		PC_13.x	PC_14.x	PC_15.x	PC_16.x	PC_17.x
## DYE2109:AAACCCACA	ACTGTGT	-1.9946774	-0.6889168	-0.8610748	-3.5271137	2.2253711
## DYE2109:AAACCCACAGGACATG		-0.1719568	-1.3127177	0.7189636	1.1575003	-
0.5746315						
## DYE2109:AAACCCACAGGCGAAT		-0.7021116	2.2996708	0.3662269	-3.1561115	1.6115634
## DYE2109:AAACCCACATCGCTAA		-2.9871921	0.3712947	-1.3384600	0.1978398	-
2.4948900						
## DYE2109:AAACCCAGTATCGGTT		-0.2702163	1.0059367	1.3182174	-1.3353218	-
0.1862197						
## DYE2109:AAACCCAGTCTGTGGC		-0.2499420	-0.3055286	-1.4799158	0.0639429	-
1.4270683						
##		PC_18.x	PC_19.x	PC_20.x	PC_21.x	
## DYE2109:AAACCCACA	ACTGTGT	-2.6493056	-1.2865859	1.17429298	-0.17077402	
## DYE2109:AAACCCACAGGACATG		0.2223031	1.7587408	0.47740384	1.28842937	
## DYE2109:AAACCCACAGGCGAAT		0.5869538	-1.7702607	-0.54852918	-0.35742542	
## DYE2109:AAACCCACATCGCTAA		-0.9557972	-0.5418771	0.30826898	0.42824778	
## DYE2109:AAACCCAGTATCGGTT		-1.9391193	0.8590362	1.50789450	-0.83889903	
## DYE2109:AAACCCAGTCTGTGGC		0.3998942	-0.4364540	0.05976991	0.03943408	
##		PC_22.x	PC_23.x	PC_24.x	PC_25.x	
## DYE2109:AAACCCACA	ACTGTGT	-0.4801928	0.3626377	-0.1410548	-2.19606867	
## DYE2109:AAACCCACAGGACATG		0.2491494	-0.3377164	-2.2398115	0.39198605	
## DYE2109:AAACCCACAGGCGAAT		0.8297882	-1.0335503	-0.8881883	1.27154928	
## DYE2109:AAACCCACATCGCTAA		0.2638503	0.5693051	1.6517613	-0.09858024	
## DYE2109:AAACCCAGTATCGGTT		0.9235641	0.2380381	1.8289060	0.33907460	
## DYE2109:AAACCCAGTCTGTGGC		-2.3471296	0.2552280	2.0036406	-0.46186112	
##		PC_26.x	PC_27.x	PC_28.x	PC_29.x	
## DYE2109:AAACCCACA	ACTGTGT	-0.79650332	-0.9703837	0.77233203	-0.88477123	
## DYE2109:AAACCCACAGGACATG		0.09935344	-2.5395412	0.01735975	0.94930022	
## DYE2109:AAACCCACAGGCGAAT		-1.51646604	-0.2892143	-0.57249556	-0.75638887	
## DYE2109:AAACCCACATCGCTAA		0.30208300	-3.7138746	1.60064143	0.05072245	
## DYE2109:AAACCCAGTATCGGTT		2.09423280	-0.3523753	-1.52177646	-0.62834183	
## DYE2109:AAACCCAGTCTGTGGC		1.18240245	-2.3649857	0.83644876	0.29585985	
##		PC_30.x	RNA_snn_res.0.10	seurat_clusters.1		
## DYE2109:AAACCCACA	ACTGTGT	0.52189970		1		4
## DYE2109:AAACCCACAGGACATG		-0.04705888		1		17
## DYE2109:AAACCCACAGGCGAAT		0.24990605		1		34
## DYE2109:AAACCCACATCGCTAA		1.06225886		1		67
## DYE2109:AAACCCAGTATCGGTT		-0.28853131		1		7
## DYE2109:AAACCCAGTCTGTGGC		2.02818241		1		51
##		RNA_snn_res.0.1.1	RNA_snn_res.0.2.1	RNA_snn_res.0.3.1		
## DYE2109:AAACCCACA	ACTGTGT		3		3	4
## DYE2109:AAACCCACAGGACATG			1		1	3
## DYE2109:AAACCCACAGGCGAAT			1		1	2

## DYE2109:AAACCCACATCGCTAA	1	3	4		
## DYE2109:AAACCCAGTATCGGTT	3	3	4		
## DYE2109:AAACCCAGTCTGTGGC	1	3	4		
##	RNA_snn_res.0.4.1	RNA_snn_res.0.5.1	RNA_snn_res.0.6.1		
## DYE2109:AAACCCACAACACTGTGT	7	8	9		
## DYE2109:AAACCCACAGGACATG	3	4	4		
## DYE2109:AAACCCACAGGCGAAT	2	2	2		
## DYE2109:AAACCCACATCGCTAA	7	3	3		
## DYE2109:AAACCCAGTATCGGTT	7	8	9		
## DYE2109:AAACCCAGTCTGTGGC	4	3	3		
##	RNA_snn_res.0.7.1	RNA_snn_res.0.8.1	RNA_snn_res.0.9.1		
## DYE2109:AAACCCACAACACTGTGT	8	10	10		
## DYE2109:AAACCCACAGGACATG	3	3	3		
## DYE2109:AAACCCACAGGCGAAT	5	5	11		
## DYE2109:AAACCCACATCGCTAA	8	6	9		
## DYE2109:AAACCCAGTATCGGTT	8	10	10		
## DYE2109:AAACCCAGTCTGTGGC	5	6	5		
##	RNA_snn_res.1.1	RNA_snn_res.2.1	RNA_snn_res.3.1		
## DYE2109:AAACCCACAACACTGTGT	10	9	5		
## DYE2109:AAACCCACAGGACATG	2	1	4		
## DYE2109:AAACCCACAGGCGAAT	11	17	20		
## DYE2109:AAACCCACATCGCTAA	9	13	28		
## DYE2109:AAACCCAGTATCGGTT	10	9	5		
## DYE2109:AAACCCAGTCTGTGGC	6	14	19		
##	RNA_snn_res.4.1	RNA_snn_res.5.1	RNA_snn_res.6.1		
## DYE2109:AAACCCACAACACTGTGT	24	22	18		
## DYE2109:AAACCCACAGGACATG	31	26	20		
## DYE2109:AAACCCACAGGCGAAT	20	34	37		
## DYE2109:AAACCCACATCGCTAA	34	31	22		
## DYE2109:AAACCCAGTATCGGTT	23	16	10		
## DYE2109:AAACCCAGTCTGTGGC	25	13	19		
##	RNA_snn_res.7.1	RNA_snn_res.8.1	RNA_snn_res.9.1		
## DYE2109:AAACCCACAACACTGTGT	14	13	12		
## DYE2109:AAACCCACAGGACATG	26	24	19		
## DYE2109:AAACCCACAGGCGAAT	35	43	34		
## DYE2109:AAACCCACATCGCTAA	19	65	65		
## DYE2109:AAACCCAGTATCGGTT	9	6	13		
## DYE2109:AAACCCAGTCTGTGGC	38	48	47		
##	RNA_snn_res.10.1	UMAP_1.y	UMAP_2.y	PC_1.y	
## DYE2109:AAACCCACAACACTGTGT	4	3.519175	-1.921958	9.663225	
## DYE2109:AAACCCACAGGACATG	17	-1.152715	3.024430	7.195771	
## DYE2109:AAACCCACAGGCGAAT	34	-2.008140	2.096559	2.055668	
## DYE2109:AAACCCACATCGCTAA	67	4.085037	1.523171	12.802924	
## DYE2109:AAACCCAGTATCGGTT	7	4.003570	-0.404878	13.135035	
## DYE2109:AAACCCAGTCTGTGGC	51	1.378143	3.335749	10.007553	
##	PC_2.y	PC_3.y	PC_4.y	PC_5.y	PC_6.y
## DYE2109:AAACCCACAACACTGTGT	7.078574	4.0789877	-0.8989851	3.4785479	4.6833617
## DYE2109:AAACCCACAGGACATG	-10.952481	-3.0915320	2.2747797	1.4447327	2.2701193
## DYE2109:AAACCCACAGGCGAAT	-12.429890	3.2396863	-2.3822854	1.1932545	-
##	2.5911117				
## DYE2109:AAACCCACATCGCTAA	1.169744	-1.3242469	-2.7640391	1.8122737	-

0.9178103

DYE2109:AAACCCAGTATCGGTT 8.753355 0.2016875 0.5418655 0.2573033 0.4390628

DYE2109:AAACCCAGTCTGTGGC -7.405388 -4.3712622 -0.8848126 -0.6372359 -
3.5383855

PC_7.y PC_8.y PC_9.y PC_10.y

DYE2109:AAACCCACAACGTGTGT -1.579191056 -1.0247210 0.81710793 -2.1728923

DYE2109:AAACCCACAGGACATG -0.277084461 0.8129334 -6.87471206 -3.9070574

DYE2109:AAACCCACAGGCGAAT 1.823911254 0.4732263 2.15869300 0.8648879

DYE2109:AAACCCACATCGCTAA 0.009797272 -0.9995096 -0.03948597 -2.4479940

DYE2109:AAACCCAGTATCGGTT -1.294562778 -1.0672499 -0.18567736 1.4473227

DYE2109:AAACCCAGTCTGTGGC 1.002155348 -0.8488276 -0.67607856 0.8396755

PC_11.y PC_12.y PC_13.y PC_14.y PC_15.y

DYE2109:AAACCCACAACGTGTGT 0.5505248 -2.481885 -1.9946774 -0.6889168 -

0.8610748

DYE2109:AAACCCACAGGACATG 0.4317886 2.132460 -0.1719568 -1.3127177 0.7189636

DYE2109:AAACCCACAGGCGAAT -2.0360958 3.149628 -0.7021116 2.2996708 0.3662269

DYE2109:AAACCCACATCGCTAA 0.1144913 1.993451 -2.9871921 0.3712947 -

1.3384600

DYE2109:AAACCCAGTATCGGTT 0.7199554 -0.365992 -0.2702163 1.0059367 1.3182174

DYE2109:AAACCCAGTCTGTGGC 1.5194082 1.726608 -0.2499420 -0.3055286 -

1.4799158

PC_16.y PC_17.y PC_18.y PC_19.y

DYE2109:AAACCCACAACGTGTGT -3.5271137 2.2253711 -2.6493056 -1.2865859

DYE2109:AAACCCACAGGACATG 1.1575003 -0.5746315 0.2223031 1.7587408

DYE2109:AAACCCACAGGCGAAT -3.1561115 1.6115634 0.5869538 -1.7702607

DYE2109:AAACCCACATCGCTAA 0.1978398 -2.4948900 -0.9557972 -0.5418771

DYE2109:AAACCCAGTATCGGTT -1.3353218 -0.1862197 -1.9391193 0.8590362

DYE2109:AAACCCAGTCTGTGGC 0.0639429 -1.4270683 0.3998942 -0.4364540

PC_20.y PC_21.y PC_22.y PC_23.y

DYE2109:AAACCCACAACGTGTGT 1.17429298 -0.17077402 -0.4801928 0.3626377

DYE2109:AAACCCACAGGACATG 0.47740384 1.28842937 0.2491494 -0.3377164

DYE2109:AAACCCACAGGCGAAT -0.54852918 -0.35742542 0.8297882 -1.0335503

DYE2109:AAACCCACATCGCTAA 0.30826898 0.42824778 0.2638503 0.5693051

DYE2109:AAACCCAGTATCGGTT 1.50789450 -0.83889903 0.9235641 0.2380381

DYE2109:AAACCCAGTCTGTGGC 0.05976991 0.03943408 -2.3471296 0.2552280

PC_24.y PC_25.y PC_26.y PC_27.y

DYE2109:AAACCCACAACGTGTGT -0.1410548 -2.19606867 -0.79650332 -0.9703837

DYE2109:AAACCCACAGGACATG -2.2398115 0.39198605 0.09935344 -2.5395412

DYE2109:AAACCCACAGGCGAAT -0.8881883 1.27154928 -1.51646604 -0.2892143

DYE2109:AAACCCACATCGCTAA 1.6517613 -0.09858024 0.30208300 -3.7138746

DYE2109:AAACCCAGTATCGGTT 1.8289060 0.33907460 2.09423280 -0.3523753

DYE2109:AAACCCAGTCTGTGGC 2.0036406 -0.46186112 1.18240245 -2.3649857

PC_28.y PC_29.y PC_30.y

DYE2109:AAACCCACAACGTGTGT 0.77233203 -0.88477123 0.52189970

DYE2109:AAACCCACAGGACATG 0.01735975 0.94930022 -0.04705888

DYE2109:AAACCCACAGGCGAAT -0.57249556 -0.75638887 0.24990605

DYE2109:AAACCCACATCGCTAA 1.60064143 0.05072245 1.06225886

DYE2109:AAACCCAGTATCGGTT -1.52177646 -0.62834183 -0.28853131

DYE2109:AAACCCAGTCTGTGGC 0.83644876 0.29585985 2.02818241

saveRDS(seurat_obj, file = paste0(results_dir, "/", glue::glue("seurat_obj_integrated_{inte

10 Session Info

```
## R version 4.4.0 (2024-04-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.4 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p-r0.3.20.so; LAPACK vers
##
## locale:
## [1] C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] knitr_1.49          reshape2_1.4.4      SeuratWrappers_0.2.0
## [4] RcppPlanc_1.0.0     rliger_2.1.0.9004   harmony_1.2.3
## [7] Rcpp_1.0.14         future_1.34.0       RColorBrewer_1.1-3
## [10] ggthemes_5.1.0      patchwork_1.3.0     scooter_0.0.0.9004
## [13] SeuratObject_5.0.2  Seurat_4.4.0        lubridate_1.9.4
## [16] forcats_1.0.0       stringr_1.5.1       dplyr_1.1.4
## [19] purrr_1.0.2         readr_2.1.5         tidyr_1.3.1
## [22] tibble_3.2.1        ggplot2_3.5.1       tidyverse_2.0.0
## [25] yaml_2.3.10
##
## loaded via a namespace (and not attached):
## [1] jsonlite_1.8.9      magrittr_2.0.3      ggbeeswarm_0.7.2
## [4] spatstat.utils_3.1-2 farver_2.1.2         rmarkdown_2.29
## [7] ragg_1.3.3          vctr_0.6.5          ROCR_1.0-11
## [10] spatstat.explore_3.3-4 tinytex_0.54         htmltools_0.5.8.1
## [13] sass_0.4.9          sctransform_0.4.1   parallelly_1.42.0
## [16] bslib_0.9.0         KernSmooth_2.23-26  htmlwidgets_1.6.4
## [19] ica_1.0-3           plyr_1.8.9          cachem_1.1.0
## [22] plotly_4.10.4       zoo_1.8-12          igraph_2.1.4
## [25] mime_0.12           lifecycle_1.0.4     pkgconfig_2.0.3
## [28] rsvd_1.0.5          Matrix_1.7-2        R6_2.5.1
## [31] fastmap_1.2.0       fitdistrplus_1.2-2  shiny_1.10.0
## [34] digest_0.6.37       colorspace_2.1-1    S4Vectors_0.44.0
## [37] tensor_1.5          irlba_2.3.5.1       textshaping_1.0.0
## [40] labeling_0.4.3      progressr_0.15.1    spatstat.sparse_3.1-0
## [43] timechange_0.3.0    httr_1.4.7          polyclip_1.10-7
## [46] abind_1.4-8         compiler_4.4.0      remotes_2.5.0
## [49] bit64_4.6.0-1       withr_3.0.2         MASS_7.3-64
## [52] ggsci_3.2.0         tools_4.4.0         vipor_0.4.7
## [55] lmtest_0.9-40       beeswarm_0.4.0      httpuv_1.6.15
```

## [58] future.apply_1.11.3	goftest_1.2-3	glue_1.8.0
## [61] nlme_3.1-167	promises_1.3.2	Rtsne_0.17
## [64] cluster_2.1.8	generics_0.1.3	gtable_0.3.6
## [67] spatstat.data_3.1-4	tzdb_0.4.0	data.table_1.16.4
## [70] hms_1.1.3	sp_2.2-0	BiocGenerics_0.52.0
## [73] spatstat.geom_3.3-5	RcppAnnoy_0.0.22	ggrepel_0.9.6
## [76] RANN_2.6.2	pillar_1.10.1	spam_2.11-1
## [79] vroom_1.6.5	later_1.4.1	splines_4.4.0
## [82] lattice_0.22-6	survival_3.8-3	bit_4.5.0.1
## [85] deldir_2.0-4	tidyselect_1.2.1	miniUI_0.1.1.1
## [88] pbapply_1.7-2	gridExtra_2.3	scattermore_1.2
## [91] RhpcBLASctl_0.23-42	stats4_4.4.0	xfun_0.50
## [94] matrixStats_1.5.0	stringi_1.8.4	lazyeval_0.2.2
## [97] evaluate_1.0.3	codetools_0.2-20	BiocManager_1.30.25
## [100] cli_3.6.3	uwot_0.2.2	xtable_1.8-4
## [103] reticulate_1.40.0	systemfonts_1.2.1	jquerylib_0.1.4
## [106] munsell_0.5.1	globals_0.16.3	spatstat.random_3.3-2
## [109] png_0.1-8	ggtrastr_1.0.2	spatstat.univar_3.1-1
## [112] parallel_4.4.0	dotCall64_1.2	listenv_0.9.1
## [115] viridisLite_0.4.2	scales_1.3.0	ggribges_0.5.6
## [118] leiden_0.4.3.1	crayon_1.5.3	rlang_1.1.5
## [121] cowplot_1.1.3		